

Network Traffic Analysis and SIEM Rules Definition

Second Project Report

Gonalo Cunha – 108352

Miguel Vila – 107276

0. Project Overview

This report analyzes non-anomalous corporate network traffic to establish baseline behavior for SIEM rule development. Using dataset 10 (data10.parquet) containing verified legitimate flows, we identified normal communication patterns, bandwidth usage, and typical destinations. We developed detection rules targeting five attack vectors: internal botnet activity, HTTPS/DNS data exfiltration, command-and-control attacks, anomalous external destinations, and suspicious external access patterns. Rules were validated against test datasets containing potential malicious activity to identify compromised devices.

The analysis utilized Python/pandas for parquet log processing and GeoLite2 databases for geographic context of external connections.

1. Normal Traffic Analysis

Baseline analysis of anomaly-free traffic is essential for detecting security threats through behavioral deviations. This chapter examines normal traffic patterns including protocol/port usage, flow distributions, IP behaviors, and server/client identification.

1.1. Network Topology Analysis and Server-Client Identification

Network analysis identified **192.168.110.0/24 as the main network with 197 unique IP addresses**. Since source IPs are exclusively local while destinations include both local and external addresses, **local destinations represent internal servers**. We identified **6 unique local destination IPs functioning as internal servers**, with services determined by their protocol:port combinations:

Servers IP Address	Flow Count	Protocol:Port	Services
192.168.110.224	42421	tcp:443 ▾	HTTPS ▾
192.168.110.230	57793	udp:53 ▾	DNS ▾
192.168.110.231	43460	tcp:443 ▾	HTTPS ▾
192.168.110.234	57451	udp:53 ▾	DNS ▾
192.168.110.235	42567	tcp:443 ▾	HTTPS ▾
192.168.110.238	42225	tcp:443 ▾	HTTPS ▾

Table 1: Internal servers and services identification.

1.2. Normal Traffic Characteristics

- **Protocol Distribution:** 88.1% TCP (849,657 flows), 11.9% UDP (115,244 flows)
- **Port Usage:** HTTPS (443) with 849,657 flows, DNS (53) with 115,244 flows
- **Normal Up/Down Ratio:** 0.104 - 0.112 (Mean: 0.108, Median: 0.109, Std: 0.002)
- **Normal HTTPS/DNS Ratio:** 6.492 - 8.442 (Mean: 7.411, Median: 7.395, Std: 0.610)
- **Up/Down Distribution:** 9.8% Upload (9.0 GB), 90.2% Download (83.4 GB)

1.3. Geographical Analysis of External Connections

Defining normally contacted countries is crucial for understanding the geographical distribution of network traffic. This helps in identifying potential security threats, such as unexpected traffic to countries that are not typically associated with the organization's operations.

- **Total countries contacted:** 37
- **Countries contacted:** AE, AR, AU, BE, BH, BR, CA, CH, CL, CN, DE, ES, FI, FR, GB, HK, ID, IE, IL, IN, IT, JP, KR, MX, MY, NL, NO, NZ, OM, PT, SA, SE, SG, TH, TW, US, ZA

Country	Percentage	Flows Ratio	Country	Percentage	Flows Ratio
United States	49.0 ▾	0.108 ▾	Spain	0.8 ▾	0.109 ▾
Portugal	41.5 ▾	0.108 ▾	Brazil	0.7 ▾	0.110 ▾
Netherlands	3.1 ▾	0.108 ▾	Ireland	0.2 ▾	0.105 ▾
Germany	2.8 ▾	0.109 ▾	India	0.1 ▾	0.113 ▾
United Kingdom	1.3 ▾	0.109 ▾	Australia	0.1 ▾	0.107 ▾

Table 2: Top 10 destination countries (99,6% of external flows).

2. Defining SIEM Rules

In this chapter, we define SIEM rules based on normal traffic patterns to detect network anomalies that may indicate attacks. We focus on five key areas: internal botnet activity through unusual device communication, data exfiltration via HTTPS and DNS channels, command-and-control communications using DNS traffic, connections to suspicious external destinations, and abnormal external user behavior when accessing corporate services. Each rule uses statistical thresholds derived from baseline data to minimize false positives while catching real threats.

2.1. Rule 1: Detecting Internal Botnet Activity

This rule identifies compromised internal devices by analyzing abnormal internal-to-internal communication patterns and detecting lateral movement activities. The detection operates on two complementary approaches:

Volume-based Detection
Baseline Average = 241.9 flows per IP pair (1182 pairs) Moderate Threshold = Baseline × 5 >= 1,209.5 flows High Threshold = Baseline × 10 >= 2,418.9 flows Critical Threshold = Baseline × 25 >= 6,047.3 flows
Lateral Movement Detection
• Server Identification: Internal servers are identified by analyzing destination IP traffic patterns in normal data • Lateral Movement Rule: Any internal-to-internal communication where the destination IP is NOT a server is flagged as lateral movement • Severity Classification: All lateral movement is automatically classified as CRITICAL regardless of flow count

2.1.1. Results Summary

Source IP	Target IPs
192.168.110.196	192.168.110.146, 192.168.110.96, 192.168.110.137
192.168.110.146	192.168.110.196, 192.168.110.96, 192.168.110.137
192.168.110.96	192.168.110.196, 192.168.110.146, 192.168.110.137
192.168.110.137	192.168.110.96, 192.168.110.196, 192.168.110.146

Table 3: Clients that made lateral connections and their targets.

The clients performing the lateral movement are communicating bidirectionally in a full mesh pattern (each compromised device communicates with all other compromised devices). **Lateral Movement Detection detected 4 clients** (192.168.110.196, 192.168.110.146, 192.168.110.96, 192.168.110.137).

From the table below, we can see that **Volume-Based Detection detected 3 critical clients** (192.168.110.21, 192.168.110.136, 192.168.110.191) communicating with internal DNS servers more than usual, indicating potential botnet activity.

Source IP	Destination IP	Flow Count	Severity	Multiplier
192.168.110.21 ▾	192.168.110.230 ▾	34,283	CRITICAL ▾	141.7×
192.168.110.21 ▾	192.168.110.234 ▾	33,327	CRITICAL ▾	137.8×
192.168.110.136 ▾	192.168.110.230 ▾	26,095	CRITICAL ▾	107.9×

192.168.110.136 ▾	192.168.110.234 ▾	25,480	CRITICAL ▾	105.3×
192.168.110.191 ▾	192.168.110.230 ▾	14,355	CRITICAL ▾	59.3×
192.168.110.191 ▾	192.168.110.234 ▾	13,547	CRITICAL ▾	56.0×

Table 4: Internal Botnet Volume-Based Detections

2.2. Rule 2: Detecting HTTPS Data Exfiltration

This rule identifies compromised internal devices by analyzing abnormal upload/download ratios in HTTPS traffic, targeting potential data exfiltration activities. The detection focuses on devices exhibiting significantly higher upload ratios compared to normal baseline behavior, which may indicate unauthorized data transmission to external destinations.

The detection operates by calculating upload/download ratios for each device's HTTPS traffic and comparing against established baseline thresholds derived from normal network behavior:

Baseline Analysis
Normal Range = [0.104, 0.112] (5th-95th percentile from baseline data) Baseline Mean = 0.108, Median = 0.108, Standard Deviation = 0.002 Boundary Tolerance = ± 0.005 (to reduce false positives)
Detection Thresholds
Moderate Threshold > 0.117 (normal max + tolerance) High Threshold > 0.559 (5× normal maximum) Critical Threshold > 1.117 (10× normal maximum)

2.2.1. Results Summary

Source IP	Ratio	Up (MB)	Down (MB)	Severity	% Above Normal
192.168.110.122	10.820	3,676.7	339.8	CRITICAL	9584.8%
192.168.110.149	7.936	2,378.0	299.6	CRITICAL	7003.6%
192.168.110.193	7.793	4,943.2	634.3	CRITICAL	6874.9%
192.168.110.185	0.366	174.8	478.0	MODERATE	227.4%
192.168.110.79	0.329	157.7	478.8	MODERATE	194.8%
192.168.110.190	0.259	97.3	375.7	MODERATE	131.9%

Table 5: HTTPS data exfiltration devices

The critical cases demonstrate clear data exfiltration patterns, with devices uploading substantially more data than they download, contrary to normal corporate network behavior where downloads typically exceed uploads by a factor of 9:1.

2.3. Rule 3: Detecting DNS Data Exfiltration

This rule identifies potential DNS-based data exfiltration and command-and-control activities by analyzing abnormal HTTPS/DNS traffic ratios. The detection focuses on devices exhibiting excessive DNS queries relative to HTTPS traffic, which may indicate DNS tunneling for covert data transmission.

The detection operates by calculating HTTPS/DNS flow ratios for each device and comparing against established baseline thresholds to identify DNS tunneling (excessive DNS usage) patterns.

Baseline Analysis
Normal HTTPS/DNS Ratio Range = [6.492, 8.442] (5th-95th percentile from baseline data) Baseline Mean = 7.411, Median = 7.395, Standard Deviation = 0.610 Analysis based on 197 devices with both HTTPS and DNS traffic
Detection Thresholds
Critical DNS Tunneling < 1.948 (excessive DNS queries) High DNS Tunneling < 4.544 (suspicious DNS activity)

2.3.1. Results Summary

Source IP	Ratio	HTTPS	DNS	Severity	% Deviation
192.168.110.191	0.078	2,169	27,902	CRITICAL	98.8%
192.168.110.21	0.110	7,461	67,610	CRITICAL	98.3%
192.168.110.136	0.135	6,984	51,575	CRITICAL	97.9%
192.168.110.171	2.461	5,206	2,115	HIGH	62.1%
192.168.110.55	3.403	5,789	1,701	HIGH	47.6%
192.168.110.88	3.500	28	8	HIGH	46.1%

2.4. Rule 4: DNS Command & Control (C&C) Detection

This rule detects potential command-and-control communications through excessive DNS query patterns. The detection identifies devices whose DNS activity is so far outside normal behavior that it indicates automated C&C communications rather than normal user activity.

Baseline Analysis
Mean: 585.0 queries per device Median: 537.0 queries per device Standard deviation: 313.7 queries (shows normal variation range) Normal range: 46 - 1,655 queries 95th percentile: 1,161.8 queries, 99th percentile: 1,543.8 queries

Detection Thresholds
High > 1,968.7 queries (max + 1 standard deviation) Critical > 3,310.0 queries (2× maximum normal)

2.4.1. Results Summary

Source IP	DNS Queries	Std Above Mean	% Above Max	Severity
192.168.110.21	67,610	213.7 std	3,985%	CRITICAL
192.168.110.136	51,575	162.5 std	3,016%	CRITICAL
192.168.110.191	27,902	87.1 std	1,586%	CRITICAL
192.168.110.171	2,115	4.9 std	27.8%	HIGH

We identified 4 suspicious devices with DNS query patterns exceeding the statistical thresholds with the most extreme case being 192.168.110.21.

2.5. Rule 5: Anomalous Countries Detection

This rule identifies suspicious external communications by analyzing geographical access patterns and detecting both new country destinations and significant traffic anomalies.

Baseline Analysis
Normal operations: 37 countries accessed during baseline period Countries contacted: AE, AR, AU, BE, BH, BR, CA, CH, CL, CN, DE, ES, FI, FR, GB, HK, ID, IE, IL, IN, IT, JP, KR, MX, MY, NL, NO, NZ, OM, PT, SA, SE, SG, TH, TW, US, ZA Total external flows: 678,984 flows across all countries
Detection Criteria
New Country Detection: Countries accessed on test day but not in baseline Significant Traffic Increases: Countries with > 700% traffic increase compared to baseline Geopolitical Sensitivity: Special focus on Iran (IR), Russia (RU), China (CN)

2.5.1. Results Summary

New Countries Detected (3 countries): **Iran (IR)**: 3 IPs, 278 flows, 43.1 MB; **Russia (RU)**: 3 IPs, 1,416 flows, 218.0 MB; **Poland (PL)**: 2 IPs, 3 flows, 0.5 MB (irrelevant).

IP Address	Russia Flows	Iran Flows	Combined Traffic (MB)
192.168.110.142	445	104	86.1
192.168.110.161	447	103	85.6
192.168.110.22	524	71	89.3

Table 6: IPs with First-Time Access to Geopolitically Sensitive Countries

Country	Baseline Flows	Test Flows	Increase	Traffic (MB)
China (CN)	34	809	+2,279%	122.0
UAE (AE)	11	57	+418%	7.3
Chile (CL)	6	18	+200%	1.9
Bahrain (BH)	60	133	+122%	12.7

Table 7: Countries with >100% traffic increases compared to baseline:

IP Address	Country	Flow Count
192.168.110.142	China	320
192.168.110.22	China	213
192.168.110.161	China	206
192.168.110.173	Bahrain	22
192.168.110.47	Bahrain	16

Table 8: Top 5 IPs Contributing to Traffic Increases

2.6. Rule 6: Suspicious External Access Patterns Detection

This rule detects automated bot behavior and anomalous access patterns from external users accessing corporate servers. The detection uses Coefficient of Variation (CV) analysis to distinguish between human and automated behavior based on timing consistency between network flows.

Statistical Terms Explanation
Coefficient of Variation (CV): Measures consistency of timing between network requests Formula: $CV = (Standard\ Deviation \div Mean) \times 100$ Low CV: Very consistent timing = automated bot, High CV: Irregular timing = likely human
Baseline Analysis
Sample size: 197 external IPs accessing corporate servers Corporate servers: 200.0.0.11 and 200.0.0.12 Mean CV: 1,164.805, Median CV: 1,281.246 Standard deviation: 540.987 Normal range: 368.983 - 1,866.845 (represents human timing variability) Average interval between flows: 9.0 seconds
Detection Criteria
Normal CV Range = [368.983, 1866.845] Critical Bot Threshold < 184.492 (50% of normal minimum) High Bot Threshold < 295.187 (80% of normal minimum) Slow Access Threshold > 27.0s (3× normal average)

2.6.1. Results Summary

IP Address	CV	Avg Int. (s)	Flows	Target	% Below Normal
176.223.2.28	23.627	8.5	3,276	200.0.0.12	93.6%
176.223.2.209	23.676	8.6	5,339	200.0.0.12	93.6%
176.223.2.116	23.751	8.6	3,262	200.0.0.12	93.6%

A CV standard deviation of 0.063 indicates that this was a coordinated attack exclusively towards target 200.0.0.12 (11,877 total flows) by 3 attackers from the same subnet (176.223.2.x). CV values of ~23.6 represent timing precision impossible for humans. **No slow access patterns detected** (all attacks used high-speed automated tools).

3. Comprehensive Threat Assessment

3.1 Summary of Detected Threats

SIEM Rule	Detections	Severity Distribution
Internal Botnet Detection	7 IPs	4 Critical (Lateral), 3 Critical (Volume)
HTTPS Data Exfiltration	6 IPs	3 Critical, 3 Moderate
DNS Data Exfiltration	6 IPs	3 Critical, 3 Moderate
DNS C&C Activities	4 IPs	3 Critical, 1 High
Anomalous Destinations	3 IPs	3 New Country, 3 Increase Country Flow
External Access Anomalies	3 IPs	3 Critical bots
Unique Malicious IPs	22 IPs	

Table 9: Threat Detection Summary

3.2 Detailed Threat Analysis

IP Address	Rules Triggered	Threat Score
192.168.110.191	Botnet Volume Crit. (4), DNS Exfil. Crit. (4), DNS C&C Crit. (4)	12
192.168.110.136	Botnet Volume Crit. (4), DNS Exfil. Crit. (4), DNS C&C Crit. (4)	12
192.168.110.21	Botnet Volume Crit. (4), DNS Exfil. Crit. (4), DNS C&C Crit. (4)	12
192.168.110.196	Botnet Lateral (4)	4
192.168.110.193	HTTPS Exfil. Cri. (4)	4
192.168.110.171	DNS Exfil. Mod. (2), DNS C&C Mod. (2)	4

192.168.110.161	Anom. Country Crit. (2) + Inc. Country Flow (2)	4
192.168.110.146	Botnet Lateral (4)	4
192.168.110.149	HTTPS Exfil. Cri. (4)	4
192.168.110.142	Anom. Country Crit. (2) + Inc. Country Flow (2)	4
192.168.110.137	Botnet Lateral (4)	4
192.168.110.122	HTTPS Exfil. Cri. (4)	4
192.168.110.96	Botnet Lateral (4)	4
192.168.110.22	Anom. Country Crit. (2) + Inc. Country Flow (2)	4
192.168.110.190	HTTPS Exfil. Mod. (2)	2
192.168.110.185	HTTPS Exfil. Mod. (2)	2
192.168.110.88	DNS Exfil. Mod. (2)	2
192.168.110.79	HTTPS Exfil. Mod. (2)	2
192.168.110.55	DNS Exfil. Mod. (2)	2

176.223.2.209	External Bot Attack: CV 23.676 (93.6% below normal), 5,339 flows to 200.0.0.12	N/A
176.223.2.116	External Bot Attack: CV 23.751 (93.6% below normal), 3,262 flows to 200.0.0.12	N/A
176.223.2.28	External Bot Attack: CV 23.627 (93.6% below normal), 3,276 flows to 200.0.0.12	N/A

Table 10: Detected IPs and causes.

4. Conclusion

The SIEM analysis successfully identified a complex multi-vector attack campaign affecting the corporate network. The detection of 22 unique malicious IP addresses (19 internal, 3 external) across six different attack vectors demonstrates both the sophistication of the threat actors and the effectiveness of the implemented detection rules.

The multi-layered approach of the SIEM rules proved essential in uncovering these sophisticated attacks that would have otherwise remained undetected through traditional security monitoring.